# Assignment 4

Part 1: Process Exploration

1. List all running processes using:

○ ps -ef

○ ps aux

2. Identify:

○ PID

○ User

○ CPU usage

○ Memory usage

```
ubuntu@ip-172-31-31-42:~$ ps -ef | head -10
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 09:33 ?        00:00:01 /sbin/init
root           2       0  0 09:33 ?        00:00:00 [kthreadd]
root           3       2  0 09:33 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 09:33 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 09:33 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 09:33 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 09:33 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 09:33 ?        00:00:00 [kworker/R-netns]
root          10       2  0 09:33 ?        00:00:00 [kworker/0:0H-events_highpri]
ubuntu@ip-172-31-31-42:~$ ps -aux | head -10
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  1.4  22192 13560 ?        Ss   09:33   0:01 /sbin/init
root           2  0.0  0.0      0     0 ?        S    09:33   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    09:33   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   09:33   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   09:33   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   09:33   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   09:33   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   09:33   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   09:33   0:00 [kworker/0:0H-events_highpri]
ubuntu@ip-172-31-31-42:~$
```

2.

1.Start a long-running command: sleep 1000 &

2. View running background jobs. – this allows us to view the running job/output in the terminal but doesn't allow us to use the terminal until the job gets finished.

3. Bring the job to foreground and send it back to background - this run behind the terminal, so it allows us to continue using the terminal

```
ubuntu@ip-172-31-31-42:~$ sleep 1000 &
[1] 7236
ubuntu@ip-172-31-31-42:~$ jobs
[1]+  Running                 sleep 1000 &
ubuntu@ip-172-31-31-42:~$ fg
sleep 1000
^Z
[1]+  Stopped                 sleep 1000
ubuntu@ip-172-31-31-42:~$ bg
[1]+ sleep 1000 &
ubuntu@ip-172-31-31-42:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-31-42:~$ ls
files  linux_lab_day1  linux_practice  logs  tmp
ubuntu@ip-172-31-31-42:~$
```

Part 3: Process Termination & Signals

1. Gracefully stop the sleep process using:

○ SIGTERM

2. Verify if the process still exists.

3. Forcefully terminate it using:

○ SIGKILL

4. Explain the difference between:

○ kill -15

○ kill -9

```
ubuntu@ip-172-31-31-42:~$ kill -15 7236
ubuntu@ip-172-31-31-42:~$ jobs
[1]+  Terminated              sleep 1000
ubuntu@ip-172-31-31-42:~$ ps aux | grep sleep
ubuntu      7310  0.0  0.2   7076  2228 pts/2    S+   16:59   0:00 grep --color=auto sleep
```

```
ubuntu@ip-172-31-31-42:~$ sleep 1000 &
[1] 7315
ubuntu@ip-172-31-31-42:~$ kill -9 7315
ubuntu@ip-172-31-31-42:~$ jobs
[1]+  Killed                  sleep 1000
ubuntu@ip-172-31-31-42:~$ ps aux | grep sleep
ubuntu      7317  0.0  0.2   7076  2224 pts/2    S+   17:04   0:00 grep --color=auto sleep
```

kill -15 (SIGTERM)

- This sends signal 15, called SIGTERM (Terminate).
- It politely asks the process to stop.
- The process gets a chance to clean up, close files, save data, and shut down safely.

kill -9 (SIGKILL)

- This sends signal 9, called SIGKILL (Kill).
- It forces the process to stop immediately, with no cleanup allowed.

Part 4: Real-Time Monitoring

1. Use top to:

○ Identify top CPU-consuming processes

○ Sort by memory usage

2. Observe changes in real time.

```
top - 17:12:44 up  7:38,  3 users,  load average: 0.00, 0.00, 0.00
Tasks: 119 total,   1 running, 114 sleeping,   4 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    914.2 total,    127.6 free,    362.3 used,    596.7 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    551.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   22192  13560   9612 S   0.0   1.4   0:01.62 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.05 ksoftirqd/0
     17 root      20   0       0      0      0 I   0.0   0.0   0:00.46 rcu_sched
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     20 root      rt   0       0      0      0 S   0.0   0.0   0:00.11 migration/0
     21 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     24 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     25 root      rt   0       0      0      0 S   0.0   0.0   0:00.15 migration/1
     26 root      20   0       0      0      0 S   0.0   0.0   0:00.06 ksoftirqd/1
     28 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
```

With sorting by memory- (shift+m)

```
top - 17:13:23 up  7:39,  3 users,  load average: 0.00, 0.00, 0.00
Tasks: 119 total,   1 running, 114 sleeping,   4 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    914.2 total,    127.6 free,    362.3 used,    596.7 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    551.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    556 root      20   0 1850044  39340  25812 S   0.0   4.2   0:02.44 snapd
    189 root      rt   0  288952  27292   8760 S   0.0   2.9   0:02.24 multipathd
    664 root      20   0  110012  23264  13808 S   0.0   2.5   0:00.12 unattended-upgr
    544 root      20   0   32416  20776  10544 S   0.0   2.2   0:00.09 networkd-dispat
    549 root      20   0 1830352  20124  11412 S   0.0   2.1   0:01.96 amazon-ssm-agen
    128 root      19  -1   66812  15636  14420 S   0.0   1.7   0:00.49 systemd-journal
    568 root      20   0  468976  13776  11484 S   0.0   1.5   0:00.65 udisksd
      1 root      20   0   22192  13560   9612 S   0.0   1.4   0:01.62 systemd
    336 systemd+  20   0   21592  12980  10652 S   0.0   1.4   0:00.14 systemd-resolve
    726 root      20   0  391876  12816  10960 S   0.0   1.4   0:00.09 ModemManager
    964 ubuntu    20   0   20096  11320   9396 S   0.0   1.2   0:00.08 systemd
   6987 root      20   0   14740  10536   8840 S   0.0   1.1   0:00.01 sshd
   6802 root      20   0   14736  10528   8828 S   0.0   1.1   0:00.01 sshd
   7108 root      20   0   14732  10516   8820 S   0.0   1.1   0:00.01 sshd
    546 polkitd   20   0  383704   9972   7544 S   0.0   1.1   0:00.62 polkitd
    491 systemd+  20   0   22408   9952   8764 S   0.0   1.1   0:00.08 systemd-network
    563 root      20   0   18152   8888   7808 S   0.0   0.9   0:00.13 systemd-logind
    192 root      20   0   26484   8356   5188 S   0.0   0.9   0:00.18 systemd-udevd
    957 root      20   0   12020   8300   7152 S   0.0   0.9   0:00.05 sshd
   7206 ubuntu    20   0   14988   7132   5120 S   0.0   0.8   0:00.18 sshd
   7043 ubuntu    20   0   14996   7112   5168 S   0.0   0.8   0:00.00 sshd
   6900 ubuntu    20   0   14992   7052   5108 S   0.0   0.8   0:00.00 sshd
   7332 ubuntu    20   0   13672   6516   4248 T   0.0   0.7   0:01.07 top
    709 syslog    20   0  222508   6316   4640 S   0.0   0.7   0:00.11 rsyslogd
   7333 ubuntu    20   0   12372   5924   3684 R   0.0   0.6   0:00.04 top
   7323 ubuntu    20   0   12372   5916   3680 T   0.0   0.6   0:00.09 top
   7330 ubuntu    20   0   12368   5876   3684 T   0.0   0.6   0:00.01 top
    536 message+  20   0    9920   5856   4788 S   0.0   0.6   0:00.53 dbus-daemon
   7207 ubuntu    20   0    9240   5568   3760 S   0.0   0.6   0:00.04 bash
   6901 ubuntu    20   0    9056   5452   3728 S   0.0   0.6   0:00.01 bash
```

To view real time and more advanced:

htop

```
 Main   I/O
  PID USER       PRI  NI  VIRT   RES   SHR S  CPU% MEM%▽  TIME+  Command
  556 root        20   0 1806M 39340 25812 S   0.0  4.2  0:01.28 /snap/snapd/current/usr/lib/snapd/snapd
  822 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.07 /snap/snapd/current/usr/lib/snapd/snapd
  823 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
  824 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.20 /snap/snapd/current/usr/lib/snapd/snapd
  825 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.00 /snap/snapd/current/usr/lib/snapd/snapd
  851 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.30 /snap/snapd/current/usr/lib/snapd/snapd
  853 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.07 /snap/snapd/current/usr/lib/snapd/snapd
  884 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.22 /snap/snapd/current/usr/lib/snapd/snapd
 5105 root        20   0 1806M 39340 25812 S   0.0  4.2  0:00.24 /snap/snapd/current/usr/lib/snapd/snapd
  189 root        RT   0  282M 27292  8760 S   0.0  2.9  0:00.73 /sbin/multipathd -d -s
  196 root        20   0  282M 27292  8760 S   0.0  2.9  0:00.00 /sbin/multipathd -d -s
  197 root        RT   0  282M 27292  8760 S   0.0  2.9  0:00.00 /sbin/multipathd -d -s
  198 root        RT   0  282M 27292  8760 S   0.0  2.9  0:00.00 /sbin/multipathd -d -s
  199 root        RT   0  282M 27292  8760 S   0.0  2.9  0:00.00 /sbin/multipathd -d -s
  200 root        RT   0  282M 27292  8760 S   0.7  2.9  0:01.52 /sbin/multipathd -d -s
  201 root        RT   0  282M 27292  8760 S   0.0  2.9  0:00.00 /sbin/multipathd -d -s
```